

Text Mining and Sentiment Analysis – Natural Language Processing

Stylometric Analysis of Writing Style in Qwen Language Models

The Aesthetic of Generation

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Motivation

Do LLMs have a stylistic “voice” of their own

- LLM research has focused heavily on fluency and factual accuracy, far less on aesthetic or stylistic properties.
- Stylometry has long identified human authors and distinguished AI writing from human writing.
- Whether different LLMs exhibit distinct styles from one another remains underexplored.

Do different variants of the same LLM family exhibit measurably distinct stylistic identities, can these be quantified through stylometric analysis?



Prior Work

Opara (2024)

Proposed the lexical/ syntactic / discourse framework this study builds on.

Kumarage et al. (2024)

Stylistic signatures identify AI-generated text even in short tweets.

Zellers et al. (2019)

Neural generation can mimic human writing convincingly, raising detection stakes.

Dataset & Pipeline

FOUR GENRES

NARRATION

An orphaned child discovers a magic mirror.

ARGUMENTATION

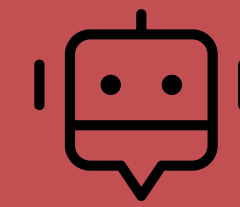
Pros and cons of AI in art and creativity.

DIALOGUE

Two old friends meet after 10 years.

DESCRIPTION

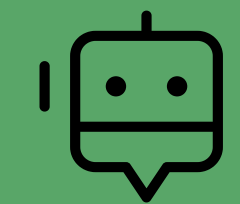
A sunset over the ocean in the Mediterranean region.



Qwen2.5-7B-Instruct

7B params – Gen 2

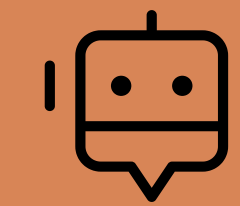
Accessed via HuggingFace Inference API - temperature = 0.7



Qwen2.5-72B-Instruct

72B params – Gen 2

Accessed via HuggingFace Inference API - temperature = 0.7



Qwen3-8B

8B params – Gen 3

Accessed via HuggingFace Inference API - temperature = 0.7

3 Models x 4 Genres x 3 Repetitions = 36 Texts

Same prompts across all models → stylistic differences are attributable to the model, not the topic.

Nine stylometric features

Lexical

Vocabulary richness and part-of-speech distribution

- Type-token ratio (lexical diversity)
- Noun Ratio
- Verb ratio
- Adjective ratio

Syntactic

Sentence complexity & structural patterns

- Average Sentence Length
- Punctuation frequency
- Syntactic Depth

Discourse

Vocabulary richness and part-of-speech distribution

- Sentiment polarity (TextBlob)
- Flesch Reading Ease (textstat)

Can a Classifier Tell the Models Apart?

01

Feature extraction

Nine stylometric features computed for each of the 36 texts.

02

Random Forest (100 trees)

Trained to predict which Qwen variant generated a given text.

03

LOO-CV Leave-On-Out CV

Trained on 35 texts, test on 1, repeated 36 times; unbiased for a small dataset.

04

Evaluate

Confusion matrix + feature importance reveal which features matter most.

Chance baseline: 33.3% (3 equally likely classes). With only 36 texts, LOO-CV squeezes maximum signal from limited data without overfitting to a fixed test split.

Average Stylometric Profile per Model

Feature	Qwen2.5-7B	Qwen2.5-72B	Qwen3-8B
Lexical Diversity	0.704	0.664	0.71
Avg Sentence Length	16.06	18.33	15.54
Adjective Ratio	0.105	0.092	0.092
Verb Ratio	0.126	0.134	0.146
Noun Ratio	0.254	0.263	0.311
Punct. Frequency	0.222	0.188	0.285
Syntactic Depth	0.086	0.093	0.075
Sentiment Polarity	0.178	0.193	0.118
Readability (Flesch)	57.63	58.51	58.47

Key characteristics

Qwen2.5-72B

Longest sentence (18.33 words), least punctuation (0.188)
→ a formal, academic register.

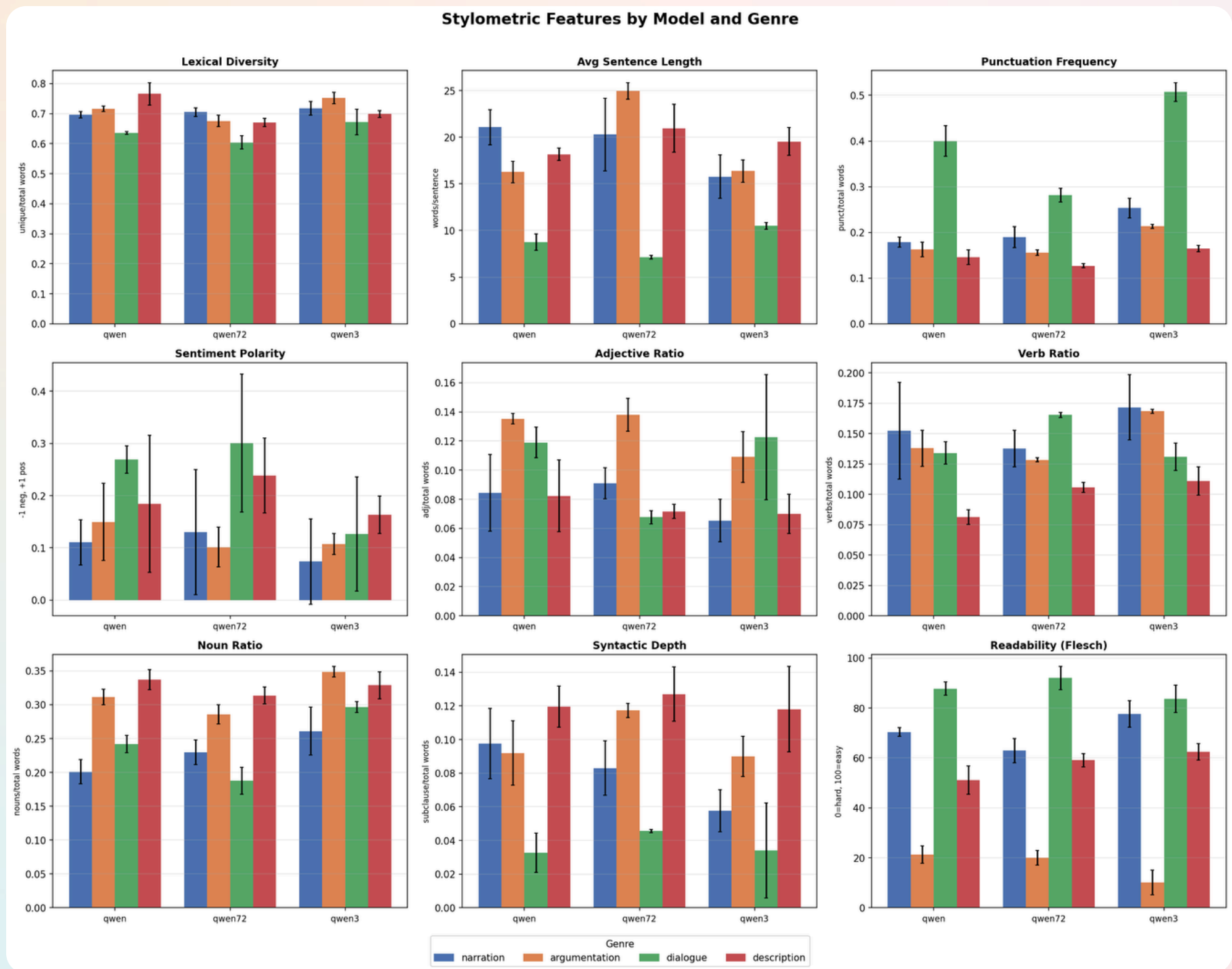
Qwen3-8B

Most punctuation overall (0.285), most pronounced in dialogue; shortest sentences on average, though longer than the others within dialogue itself.

Qwen2.5-7B

Balanced on sentence-level features, but the most extreme on part-speech ratios (highest adjective, lowest verb and noun ratio).

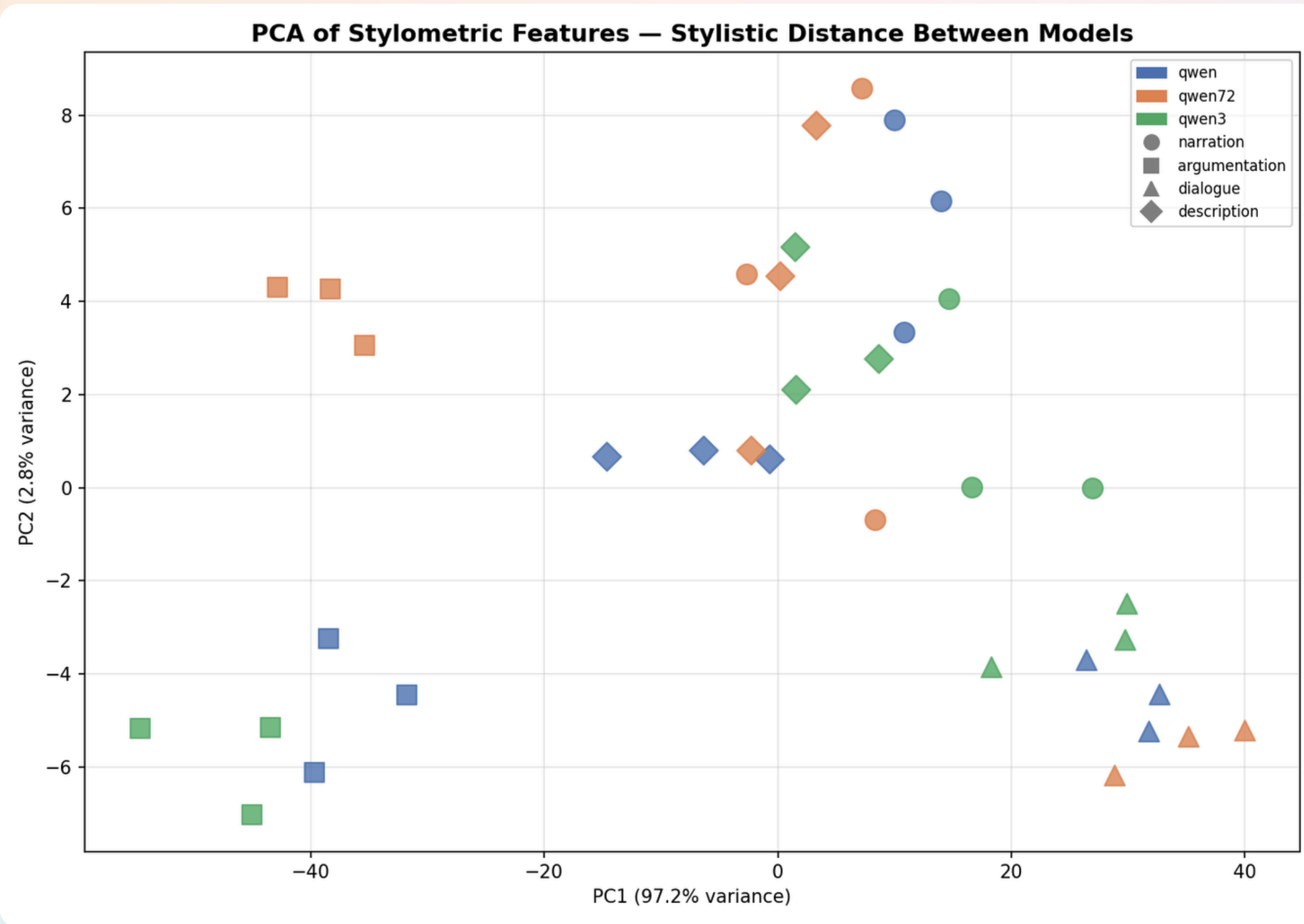
All nine features by Model and Genre



Key Findings

- Dialogue has highest punctuation, lowest sentence length, in every model.
- Lexical diversity peaks in a different genre per model: description (Qwen2.5-7B), narration (Qwen2.5-72B), argumentation (Qwen3-8B)
- Syntactic depth follows the same genre pattern: lowest in dialogue, highest in description, across all models; Qwen3-8B's narration is a notable outlier, roughly half the depth of the other two (~0.057 vs ~0.095).
- Sentiment polarity varies most in dialogue and description; Qwen2.5-7B and Qwen2.5-72B skew more positive than Qwen3-8B there.
- Readability: dialogue highest for all models; argumentation lowest.

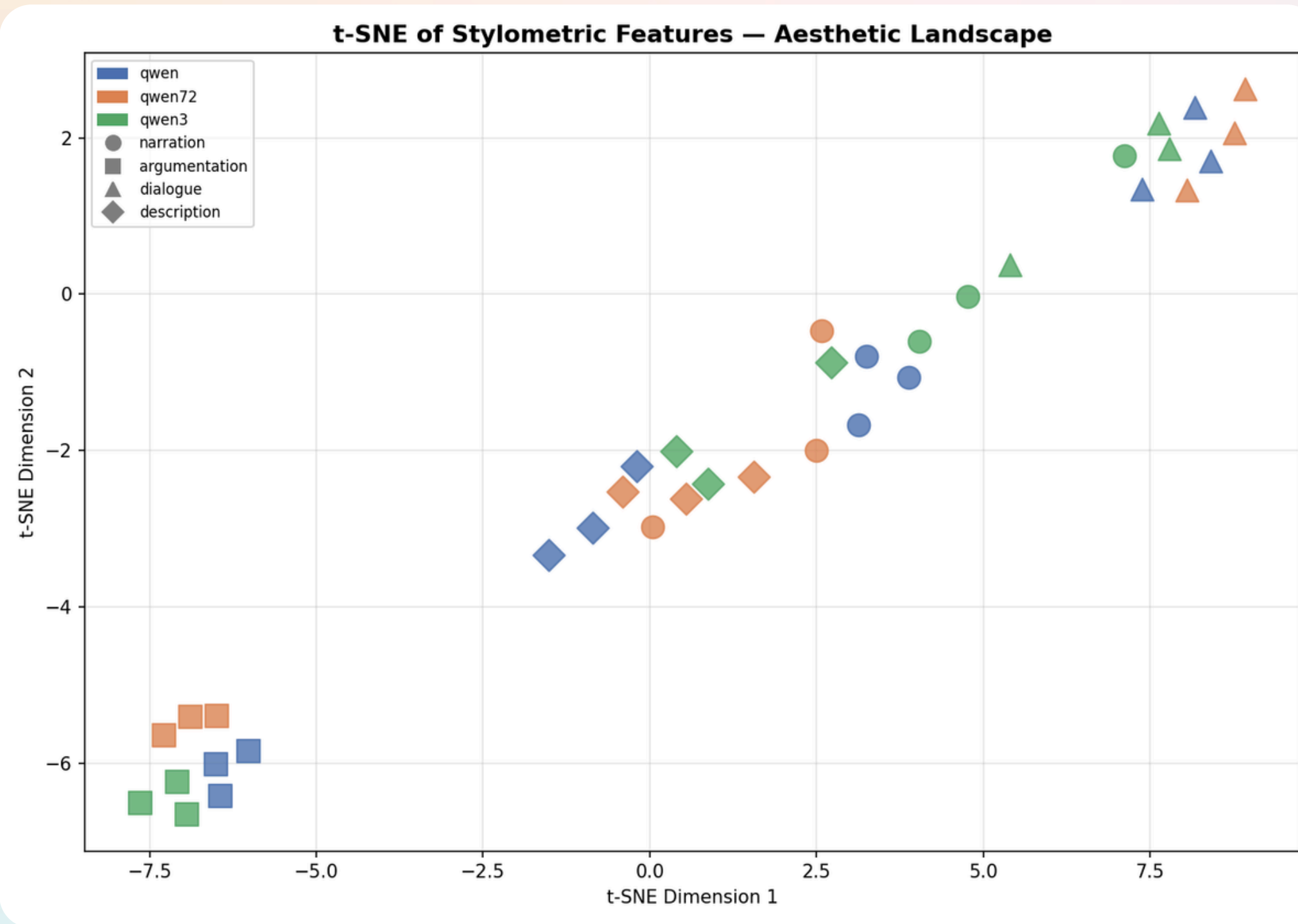
PCA: Stylistic Distance Between Texts



Key Findings

- Genre has a stronger separating effect than model identity.
- Argumentation (squares) forms the most distinct cluster, far left (PC1 negative).
- Dialogue (triangles) clusters on the far right (PC1 positive).
- Description (diamonds) sits centrally, and narration (circles) spreads across the middle and the right.
- Within each genre cluster, the three model still show partial separation, a secondary but detectable signal.

t-SNE: A Clearer Genre Landscape



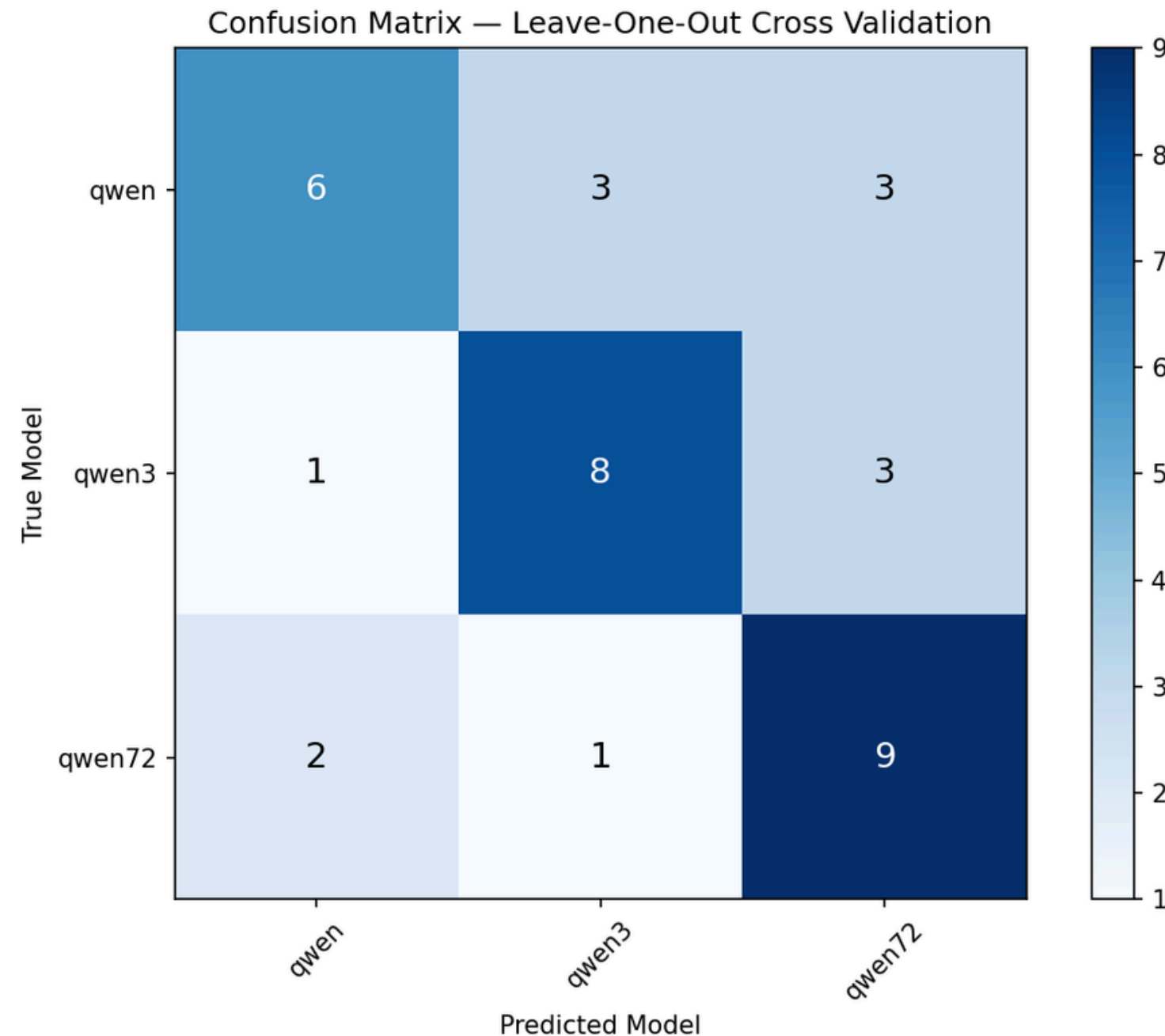
Key Findings

- Argumentation and dialogue form the tightest, most isolated cluster (lower left).
- Narration is the most scattered genre: it spreads across the center, overlapping with both description and dialogue.
- Within the argumentation, the three models show a visible colour ordering: within dialogue, they're more interspersed.
- With only 9 texts per genre, this is a tentative visual pattern; it does not favour dialogue over argumentation for detecting model identity.

Classification Results

63.9%

vs 33.3% chance baseline



Qwen2.5-7B

75%

Qwen2.5-72B

67%

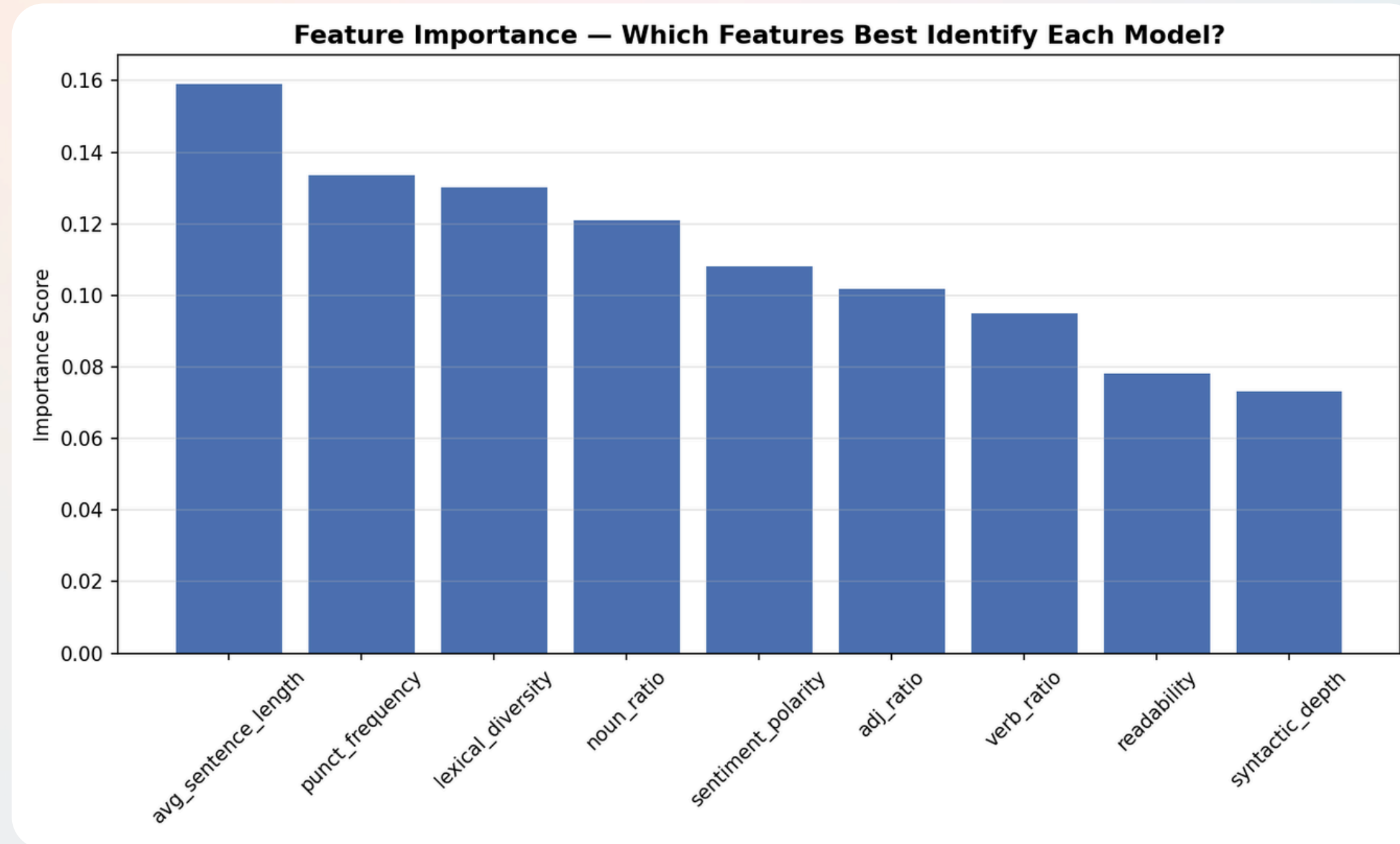
Qwen3-8B

50%

Qwen2.5-7B is confused equally with both other models (3 each), while **Qwen2.5-72B** is rarely confused with **Qwen3-8B** (only 1).

→ *Model generation differentiates style more strongly than size alone.*

Which features best identify each model?



Sentence length, **punctuation**, and **lexical diversity** carry the most discriminative signal; **readability** and **syntactic depth** carry the least.

Can Style be transferred on demand?

Research question: Can Qwen2.5-7B emulate Qwen2.5-72B's stylistic fingerprint when given explicit style instructions?

📄 Genre: NARRATION (3 runs averaged)

Feature	qwen	qwen rewrite	qwen72 target	
lexical_diversity	0.698	0.695	0.705	✗
avg_sentence_length	18.714	22.857	20.278	✗
adj_ratio	0.061	0.085	0.091	✓
verb_ratio	0.16	0.138	0.138	✓
noun_ratio	0.176	0.231	0.23	✓
punct_frequency	0.191	0.155	0.189	✗
syntactic_depth	0.122	0.123	0.083	✗
sentiment_polarity	0.168	0.058	0.13	✗
readability	69.354	34.92	62.937	✗

→ 3/9 features shifted toward qwen72 style

📄 Genre: ARGUMENTATION (3 runs averaged)

Feature	qwen	qwen rewrite	qwen72 target	
lexical_diversity	0.717	0.689	0.676	✓
avg_sentence_length	17.833	30.486	24.956	✓
adj_ratio	0.131	0.154	0.138	✗
verb_ratio	0.15	0.123	0.128	✓
noun_ratio	0.327	0.275	0.286	✓
punct_frequency	0.14	0.127	0.155	✗
syntactic_depth	0.065	0.128	0.117	✓
sentiment_polarity	0.194	0.152	0.101	✓
readability	19.59	-13.475	20.024	✗

→ 6/9 features shifted toward qwen72 style

📄 Genre: DIALOGUE (3 runs averaged)

Feature	qwen	qwen rewrite	qwen72 target	
lexical_diversity	0.631	0.696	0.604	✗
avg_sentence_length	9.947	18.455	7.14	✗
adj_ratio	0.116	0.115	0.068	✓
verb_ratio	0.127	0.156	0.165	✓
noun_ratio	0.233	0.189	0.188	✓
punct_frequency	0.354	0.132	0.282	✗
syntactic_depth	0.048	0.094	0.046	✗
sentiment_polarity	0.24	0.199	0.301	✗
readability	84.044	35.845	91.998	✗

→ 3/9 features shifted toward qwen72 style

📄 Genre: DESCRIPTION (3 runs averaged)

Feature	qwen	qwen rewrite	qwen72 target	
lexical_diversity	0.815	0.829	0.67	✗
avg_sentence_length	19.0	22.333	20.952	✓
adj_ratio	0.116	0.134	0.072	✗
verb_ratio	0.084	0.105	0.106	✓
noun_ratio	0.316	0.295	0.314	✗
punct_frequency	0.147	0.146	0.127	✓
syntactic_depth	0.105	0.107	0.127	✓
sentiment_polarity	0.297	0.166	0.238	✗
readability	43.687	17.227	59.095	✗

→ 4/9 features shifted toward qwen72 style

Can Style be transferred on demand?

Research question: Can Qwen2.5-7B emulate Qwen2.5-72B's stylistic fingerprint when given explicit style instructions?

Genre	Featured Shifted	Success Rate
Narration	3/9	33%
Argumentation	6/9	67%
Dialogue	3/9	33%
Description	4/9	44%
Overall	16/36	44.40%

44.40% of features overall shifted toward the Qwen2.5-72B profile (16/36)

- Argumentation transferred best.
- Narration and dialogue transferred least
- Lexical features were most malleable
- Readability & punctuation frequency resisted instruction: word choice shifts more easily than deeper syntactic and discourse-level style

Concluding Remarks

Qwen model variants exhibit measurably distinct stylometric fingerprints

01

Genre dominates style

Genre has a stronger effect on stylometric features than model identity (clear clustering in PCA & t-SNE)

02

Size shapes formality

Qwen2.5-72B adopts the most formal register: lower sentences, less punctuation.

03

Generation shapes expressiveness

Qwen3-8B is the most expressive overall, especially in dialogue punctuation; generation matters beyond scale.

04

Separation depends on Genre

Model identity separates most clearly within argumentation and dialogue in t-SNE, but with only nine texts per genre this is a qualitative pattern, not a confirmed result.